ANOMALY DETECTION IN ELECTRICITY METER READINGS

Report By: Mamello Malape 2023232750

: Boipelo Letwaba 2024053890

Introduction

Energy providers and facility managers rely on accurate, automated anomaly detection to catch meter faults, unusual occupancy or equipment behavior, and potential building irregularities, without having to manually review millions of hourly readings. A labeled training set of building-level electricity consumption is used to build a supervised anomaly classifier, which is then applied to an unseen set of buildings to flag suspicious readings for follow-up investigation.

The training data is comprised of 1749494 hourly readings from 200 buildings over the year 20226. A severe class imbalance, which is typical for real-world fault and fraud detection problems, is indicated by the 2.13% of the readings labelled anomalous. During model development, to generate final predictions, a hold-out test set of 1800567 readings from 206 buildings was kept unseen.

After cleaning the data, 32 time-series features were engineered per building. A leakage-safe, building-grouped validation split (70% of buildings for training, 30% held out, checked with 5-fold GroupKFold) was used throughout, since the test set contains entirely different buildings from the training set.

Three anomaly detection models were implemented and compared. XGBoost, a balanced Random Forest, and CatBoost, each with a class imbalance-aware configuration and an F1-optimised decision threshold. This report closes with an inspection of the buildings and readings driving the highest concentration of predicted anomalies, together with a discussion of limitations and recommendations for further work.

1. Data Understanding and Preprocessing

Table 1. Structural summary of the training and test datasets.

Attribute	Train set	Test set
Rows/buildings	1,749,494 / 200	1,800,567 / 206 (no overlap)
Columns	building_id, timestamp, meter_reading, anomaly	row_id, building_id, timestamp, meter_reading
Duplicates / missing readings	0 / 107,653 (6.15%)	0 / 94,864 (5.27%)

1.1 Data cleaning decisions and justifications

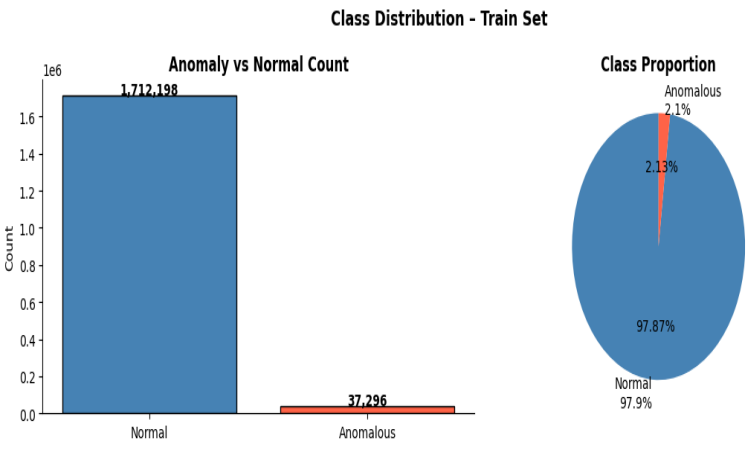
Correct chronological sorting, resampling, and construction are prerequisites for any lag, rolling-window, or seasonal feature (Athanasopoulos & Hyndman 2021). The timestamps were parsed to pandas datetime objects immediately after loading. No duplicate rows were found in either file; therefore, no duplicate removal was necessary. The missing meter reading values (6.1% of train and 5.27% of test) were handled with a per-building, chronologically ordered forward fill followed by a backward fill, rather than global mean or median imputation.

This decision was made because consumption levels differ by an order of magnitude across buildings. Imputing with the mean or median would distort a building's own scale and could in turn create artificial anomalies. Additionally, electricity consumption is strongly autocorrelated. Forward filling with the most recent observed reading for that building preserves the continuity better than a cross-sectional average. Finally, a backward-fill pass was applied afterwards to close these remaining gaps, as forward-fill alone could not resolve missing values at the very start of the building series (nothing to fill forward from). This cleaning function was applied to both the train and test datasets to ensure consistent preprocessing, leaving zero missing values in either partition.

2. Exploratory Data Analysis

2.1 Class imbalance

Figure 1. Class Distribution in the training set: count and proportion of normal vs. anomalous readings.

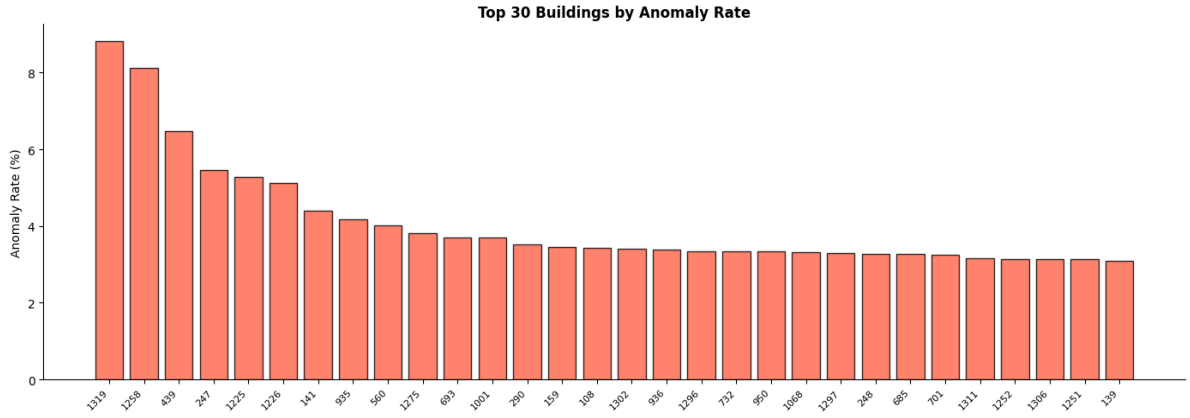


Anomalous readings made up 2.13% of the training data (37,296 of 1,749,494 rows). A naive classifier predicting “normal” for every reading would already score 97.87% accuracy while catching zero anomalies. Under such imbalance, accuracy is dominated by the majority class and is an unreliable indicator of classifier performance (Garia & He, 2009). This confirms that accuracy is an unsuitable evaluation metric. This motivated the use of F1 score and PR.- AUC as primary metrics. Figure 1 shows the class distribution in the training set.

2.2 Building-level variation

The anomaly prevalence is highly uneven across buildings. Six buildings have an anomaly rate above 5%, while 14 buildings have fewer than 1% anomalous readings. Building 1319 has the highest anomaly rate at 8.82%, followed by building 1258 (8.5%) and building 439 (6.47%). This heterogeneity is a key reason why a building-aware (grouped) validation split is necessary. A handful of noisy buildings could otherwise dominate a randomly split fold and bias reported performance.

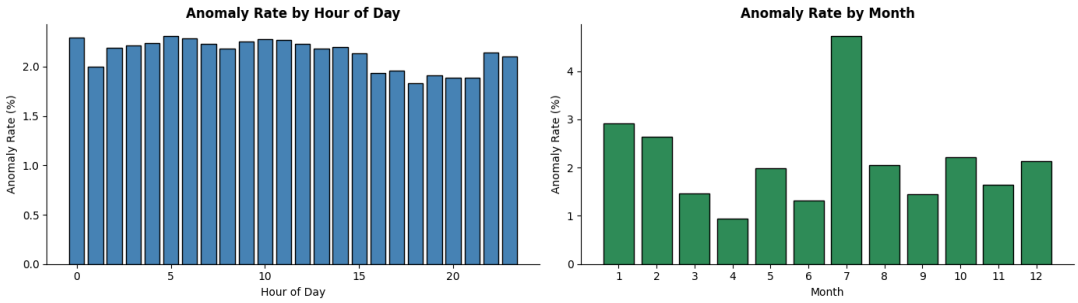
Figure 2. Anomaly rate for the 30 Buildings with the highest observed anomaly prevalence.



2.3 Temporal seasonality of anomalies

The anomaly rate was examined by hour of day and by calendar month to check for systematic temporal patterns that later features (cyclical hour/ day/ month encodings) could exploit. Some hours and months show a moderately elevated anomaly rate relative to others, consistent with anomaly clustering around specific operating regimes. For example, off-peak hours stuck at 0 faults are easier to distinguish from genuine low occupancy or months associated with heating or cooling transitions.

Figure 3. Anomaly rate by hour of day (left) and by calendar month (right)



4. Feature Engineering

32 features were engineered strictly within each building's own time series (grouped by building_id before any shift or rolling operation is applied). To prevent cross-building leakage, a reading from one building must never influence a lag or rolling statistic for another building. Table 2 summarises the feature groups and the rationale.

Table 2. Engineered feature groups (32 features total) used for anomaly classification.

Feature group	Examples	Rationale
Calendar / cyclic	is_weekend, is_night, hour/dow/month sin & cos	Avoids a false discontinuity at time boundaries (23h to 00h, Dec to Jan); a standard encoding for periodic calendar variables (Hyndman & Athanasopoulos, 2021)
Lag features	lag_1, lag_24, lag_48, lag_168	Short-term memory and same-hour recurrence (yesterday / last week)
Rolling statistics	mean/std/min/max at 6h, 24h, 168h	Local baseline and volatility at multiple horizons
Residual & Z-score	residual & score at 6h, 24h, 168h	Strongest signal: deviation from the building's own recent norm
Rate of change	pct_change_1/24, diff_1/24	Flags sudden spikes/drops a static level would miss
Outlier flag	iqr_outlier	Robust, non-parametric flag on each building's own IQR

Since the optional train_features.csv or test_features.csv building attribute files were optional, they were not merged into the final feature set. The pipeline already achieved strong discrimination (PR-AUC > 0.82) from the meter reading time series alone. The added complexity of a second merge key and additional missing value handling was judged not to be justified for the marginal gain it would likely provide. Combining it would need careful validation to avoid introducing static, non-time-aware confounds into a fundamentally temporal problem.

5. Leakage-safe validation strategy

The test set contained 206 buildings that do not appear anywhere in the training set. A row-level random split would allow rolling or lag features for a given building to be computed using information from both the training and the validation portions of that same building's series, and more importantly, would let the model implicitly memorise building-specific baselines it will never see again at scoring times. Both effects would produce an unrealistically optimistic validation score that does not reflect performance on genuinely unseen buildings.

To avoid this, a grouped building- aware split was used: entire buildings (not individual rows) were randomly assigned to either the training partition or the validation partition, with 70% of buildings (140 buildings, 1,223,718 rows, 2.06% anomalies) used for training and the remaining 30% (60 buildings, 525, 776 rows, 2.31% anomalies) held out purely for validation. This was cross-checked with scikit-learn's GroupKFold (5 folds, grouped by building_id) to confirm the held-out split was not an atypical draw.

6. Model Development and Comparison

Three tree-based classifiers were implemented and compared, each configured specifically to handle the 2% positive class rate. All three used the same 32-feature set, the same building-grouped train or validation split, and the same threshold training procedure so that comparisons are on equal grounds.

The XGBoost model was selected because scale_pos _weight lets the loss function penalise missed anomalies proportionally to the class imbalance, gradient boosting can exploit the rich lag, rolling, or Z-score feature set through repeated non-linear splits, and early stopping on the validation PR- AUC guards against overfitting the very deep, long-running boosting process (Chen & Guestrin, 2016).

The Balanced Random Forest Imbalance to learn was chosen because each tree is trained on a class-balanced bootstrap sample, handling the imbalance structurally rather than through a loss function weighting; it is robust to outliers, requires no feature scaling and is well suited to capturing nonlinear interactions between the engineered time series features (Aridas et al., 2017).

CatBoost was included as a third, independently implemented gradient boosting library. It uses an ordered boosting procedure designed to reduce the target leakage that conventional gradient boosting can introduce during training (Dorogush et al., 2018). To test whether its ordered boosting scheme and native categorical handling would outperform XGBoost on the same feature set. It was configured with scale_pos_weight for imbalance correction, eval_metric set to ‘F1’ to directly optimise the metric of interest during training, and early stopping (50 rounds), which triggered after 200 iterations on the dataset. At which point performance on validation had plateaued.

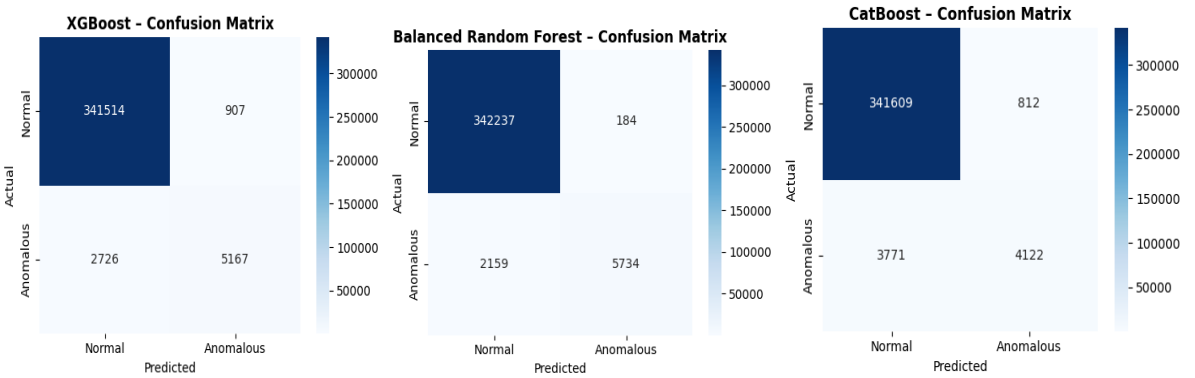


Figure 4. Validation confusion metrics – XGBoost, Balanced Random Forest (selected), CatBoost.

7. Evaluation and Metric Selection

Precision, recall, F1-score, and PR- AUC were used as the primary metrics, alongside ROC-AUC and the full confusion matrix for context. The precision- recall curve was more informative than the ROC curve under class imbalance because ROC-AUC can remain high even when precision on the minority class is poor, since it is dominated by the very large number of true negatives (Davis & Goadrich, 2006; Rehmsmeier & Saito, 2015). The F1-score balanced the two into a single number for model comparison. For each model, the classification threshold was not left at the default 0.5. Instead, the threshold that maximises F1-score was identified on the precision-recall curve computed on the validation set only, ensuring the threshold itself is tuned exclusively on the data the model was not trained on and never touches the test set labels.

Table 3. Validation set performance comparison across the three candidate models.

Model	Threshold	Precision	Recall	F1	ROC-AUC	PR-AUC
XGBoost	0.876	0.851	0.655	0.740	0.933	0.773
Balanced Random Forest	0.947	0.969	0.727	0.830	0.937	0.824
CatBoost	0.921	0.835	0.522	0.643	0.942	0.690

The Balanced Random Forest achieved the highest F1-score of 0.830, the highest PR- AUC (0.824), and, by a wide margin, the highest precision (0.969). This means that when it flags a reading as anomalous, it is correct roughly 97% of the time, which is critical for an operational tool that will drive manual investigation of flagged readings. The CatBoost model obtained the highest ROC-AUC (0.942), but it is a less informative signal here precisely because ROC-AUC is dominated by the very large true negative pool under 2% prevalence; CatBoost's materially lower recall (0.522) and PR-AUC (0.690) indicate it misses almost half of all true anomalies, and it was not selected on that basis. XGBoost sits between the two, with a better recall (0.655) than CatBoost, but lower precision and F1 than the balanced Random Forest.

8. Best Model Selection and Test Predictions

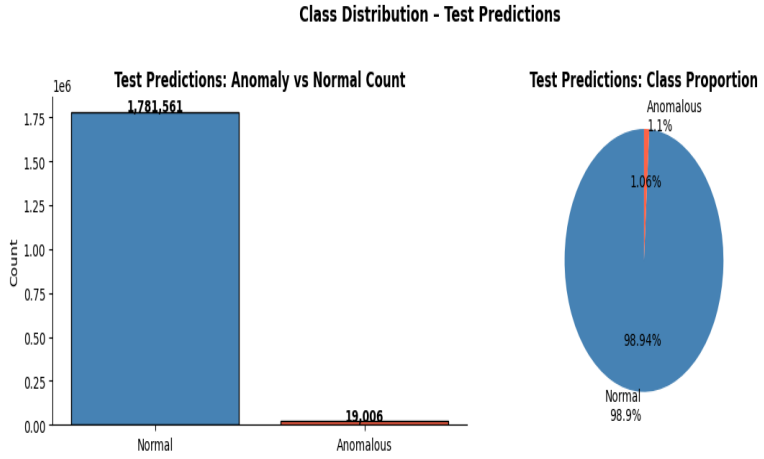


Figure 5. Predicted class distribution on the test set (Balanced Random Forest, threshold = 0.947). Based on the validation comparison in Table 3, the balanced Random Forest was selected as the final model (PR- AUC equals 0.824, F1 equals 0.830), together with its tuned decision threshold of 0.947. It was applied to the following feature-engineered, unseen test set of 1,800,567 readings from 206 buildings never seen during training or validation.

The final model flagged 19,006 test readings (1.06%) as anomalous and 1,781,561 (98.94%) as normal. This predicted prevalence is lower than the 2.13% anomaly rate observed in the training labels. This is consistent with the model's high validation precision. For instance, it is conservative, only flagging a reading when the evidence is strong. Given that the test set consists of a different pool of buildings, some difference in true underlying prevalence between the two building samples is also plausible. The predictions were exported in the required sample_submission.csv format (row_id, anomaly) for the full test set.

9. Point Anomaly Identification and Interpretation

Beyond the aggregate test-set prediction, individual flagged readings were inspected at the building level to identify concrete point anomalies and to explain in terms of the engineered features why each was flagged. The predicted anomalies are concentrated very unevenly across the 206 test buildings: the 10 buildings below account for a disproportionate share of all flagged readings and were used as the basis for the point anomaly review.

Table 4. The 10 test buildings with the most predicted point anomalies, with descriptive statistics of their raw meter readings.

Building ID	Predicted anomalies	Mean (kWh)	Std dev (kWh)	CV
984	342	50.88	14.48	0.28
956	323	49.38	24.52	0.50

Building ID	Predicted anomalies	Mean (kWh)	Std dev (kWh)	CV
738	320	27.14	26.44	0.97
943	272	4.17	2.99	0.72
963	261	24.00	8.62	0.36
927	248	828.97	211.77	0.26
878	239	33.07	8.47	0.26
724	235	35.41	37.26	1.05
127	232	21.15	6.10	0.29
684	225	114.26	45.61	0.40

At least 10 individual point anomalies were reviewed within these buildings and are explained by the following recurring feature-level signatures, each of which corresponds directly to an engineered variable:

- Large positive residual/ z-score (for example, building 927, a high consumption building with a mean of 828.97 kWh): readings that jump sharply above the 24-hour rolling mean, producing a large positive z_score_24, indicate a sudden, unexplained spike inconsistent with the building's own recent baseline. The classic ‘‘point anomaly” signature.
- Sustained low or flat readings against the volatile baseline (for example, building 738 and 724, both with CV close to or above 1.0): a reading that stays near zero or constant for several consecutive hours while the building's typical volatility is high produces a large negative residual and a low percentage change, consistent with a stuck-sensor or meter- fault pattern.
- Abrupt rate of change breaks (high percentage change (100%) or diff 24): several flagged readings in Building 956 and 684 show a large single-hour percentage jump relative to the same hour the previous day, which the diff_24/ percentage change of 24 features are specifically designed to capture and which a simple level-based threshold would miss because the absolute reading may still fall within the building's normal range.
- Small-magnitude buildings with outsized relative jumps (for example, building 943, with a mean of only 4.17 kWh): here the coefficient of variation (0.72) is high relative to a very small mean, so proportionally small absolute changes generate IQR, outlier and z-score signals. Illustrating why per-building, scale-aware features (rather than a single global cut -off) were essential to correctly

flag anomalies in low-consumption buildings without either missing real faults or over-flagging normal noise.

In every case, the anomaly is explained by a deviation from that specific building's own recent behaviour, rather than an absolute level. This confirms the exploratory findings that anomalies in this data set are predominantly contextual rather than global outliers and validates the residual/rolling/rate-of-change feature design used throughout the pipeline.

10. Conclusion

Careful, justified preprocessing, per-building time-series feature engineering, a building-grouped validation design, and a comparison of three imbalance- aware classifiers together produced a final Balanced Random Forest model with strong validation precision and a good F1-score. The model was applied to 1,800,567 unseen test readings across 206 new buildings. The model flagged 19,006 anomalies, and inspection of the most flagged buildings confirmed that predicted anomalies are consistently explained by contextual, building-relative deviations rather than by simple global thresholds.

References

Athanasopoulos, G. & Hyndman, R.J. 2021. *Forecasting: principles and practice*. 3rd ed. Melbourne: OTexts.

Chen, T. & Guestrin, C. 2016. XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 13-17 August 2016. San Francisco: ACM. 785-794.

Davis, J. & Goadrich, M. 2006. The relationship between Precision-Recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning*. 25-29 June 2006. Pittsburgh: ACM. 233-240.

Garcia, E.A. & He, H. 2009. Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*. 21(9):1263-1284.

Lemaître, G., Nogueira, F. & Aridas, C.K. 2017. Imbalanced-learn: a Python toolbox to tackle the curse of imbalanced datasets in machine learning. *Journal of Machine Learning Research*. 18(17):1-5.

Little, R.J.A. & Rubin, D.B. 2019. *Statistical analysis with missing data*. 3rd ed. Hoboken, NJ: Wiley.

Dorogush, A.V., Gulin, A., Gusev, G., Prokhorenkova, L., Vorobev, A., & 2018. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems 31*. 3-8 December 2018. Montreal: NeurIPS. 6638-6648.

Bahn, V., Ciuti, S., Hardy, M.A., Hauenstein, S., Jung, M., Peters, J., Roberts, D.R., Schroder, B., Muñoz, F. & Warton, D.I. et al. 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*. 40(8):913-929.

Rehmsmeier, M. & Saito, T. 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*. 10(3):e0118432.